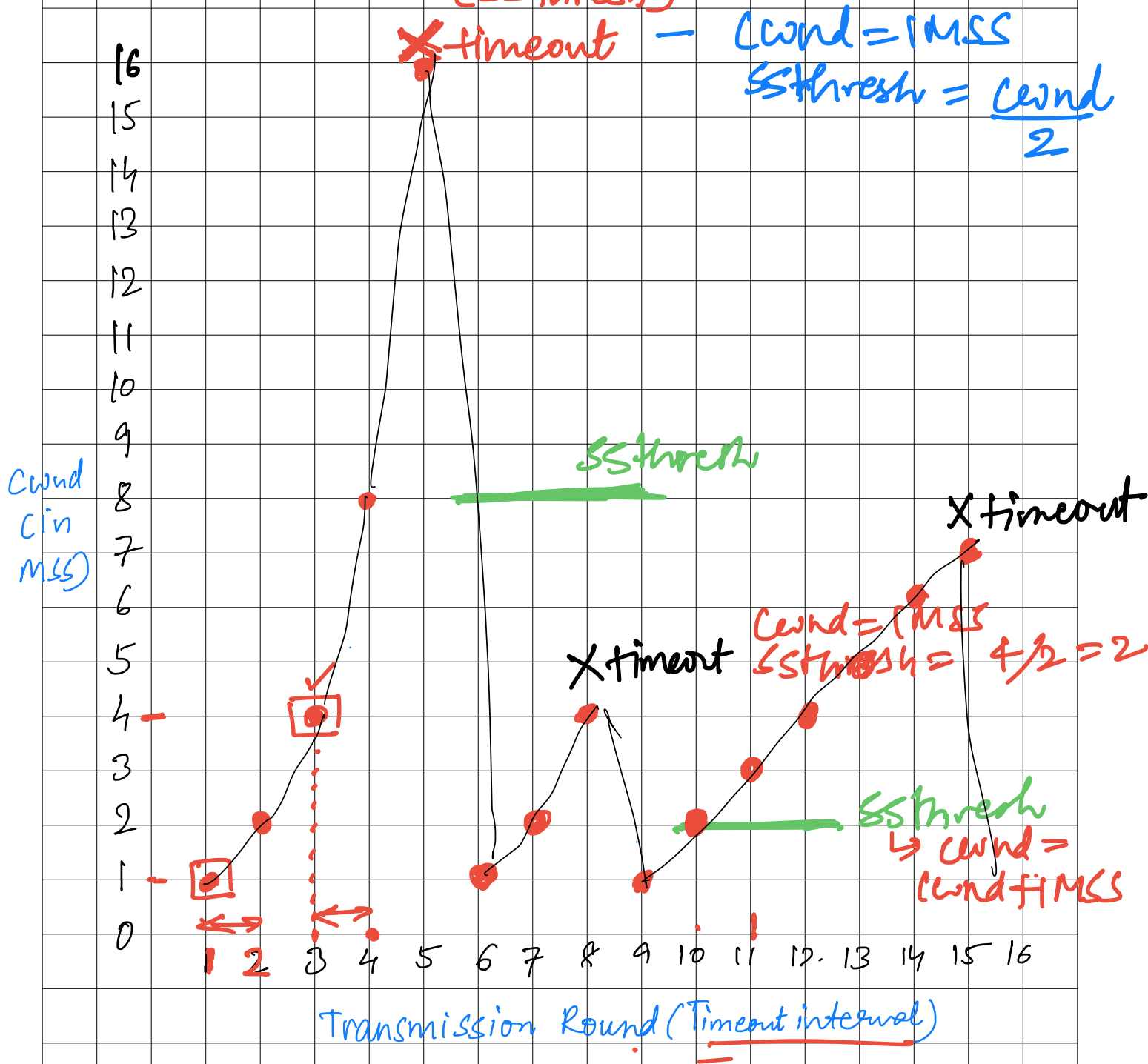
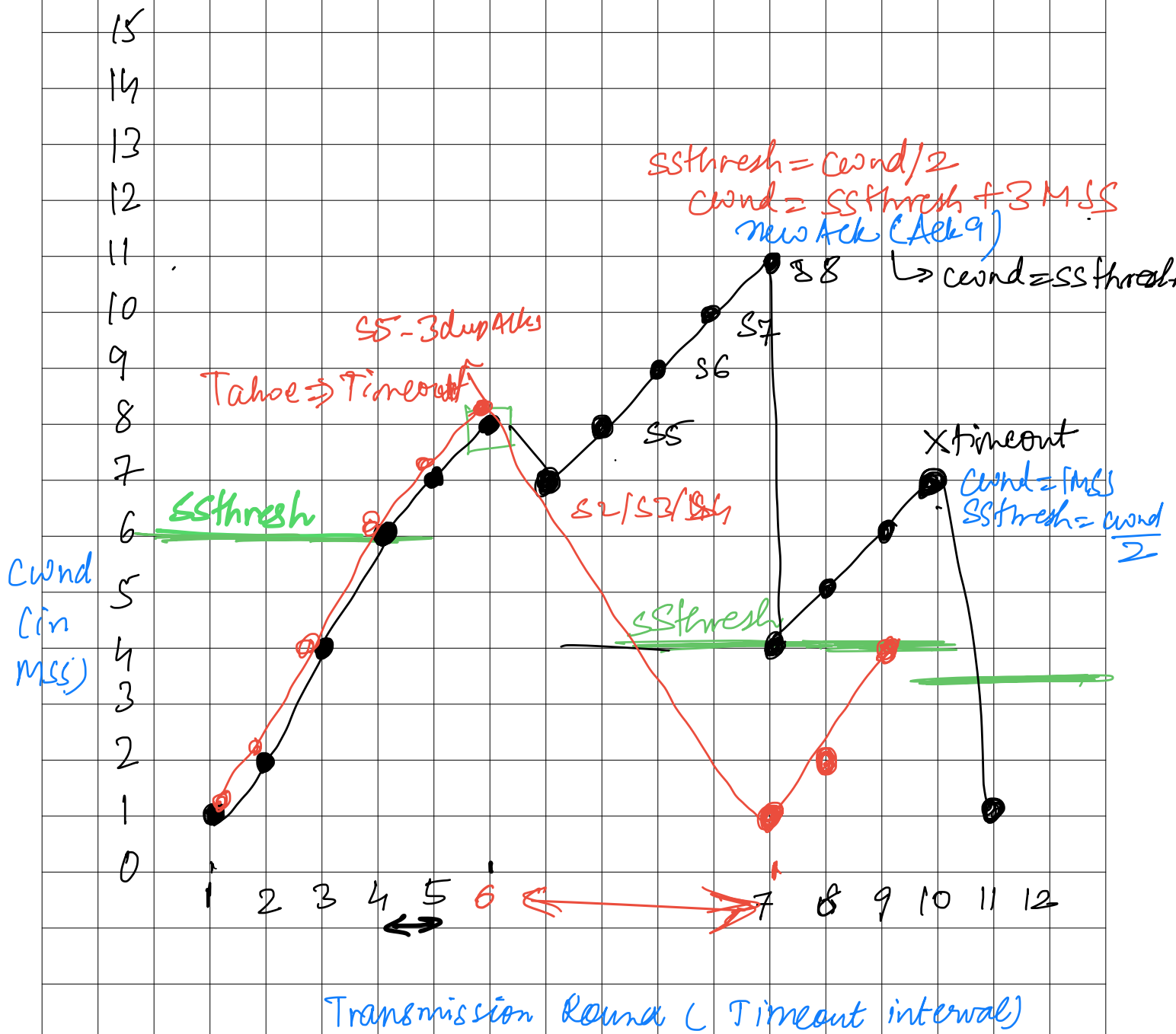


Slow start: $Cwnd = 1MSS$ Ack ✓ double $cwnd$
 Congestion avoidance
 Slow start threshold (ss_thresh)

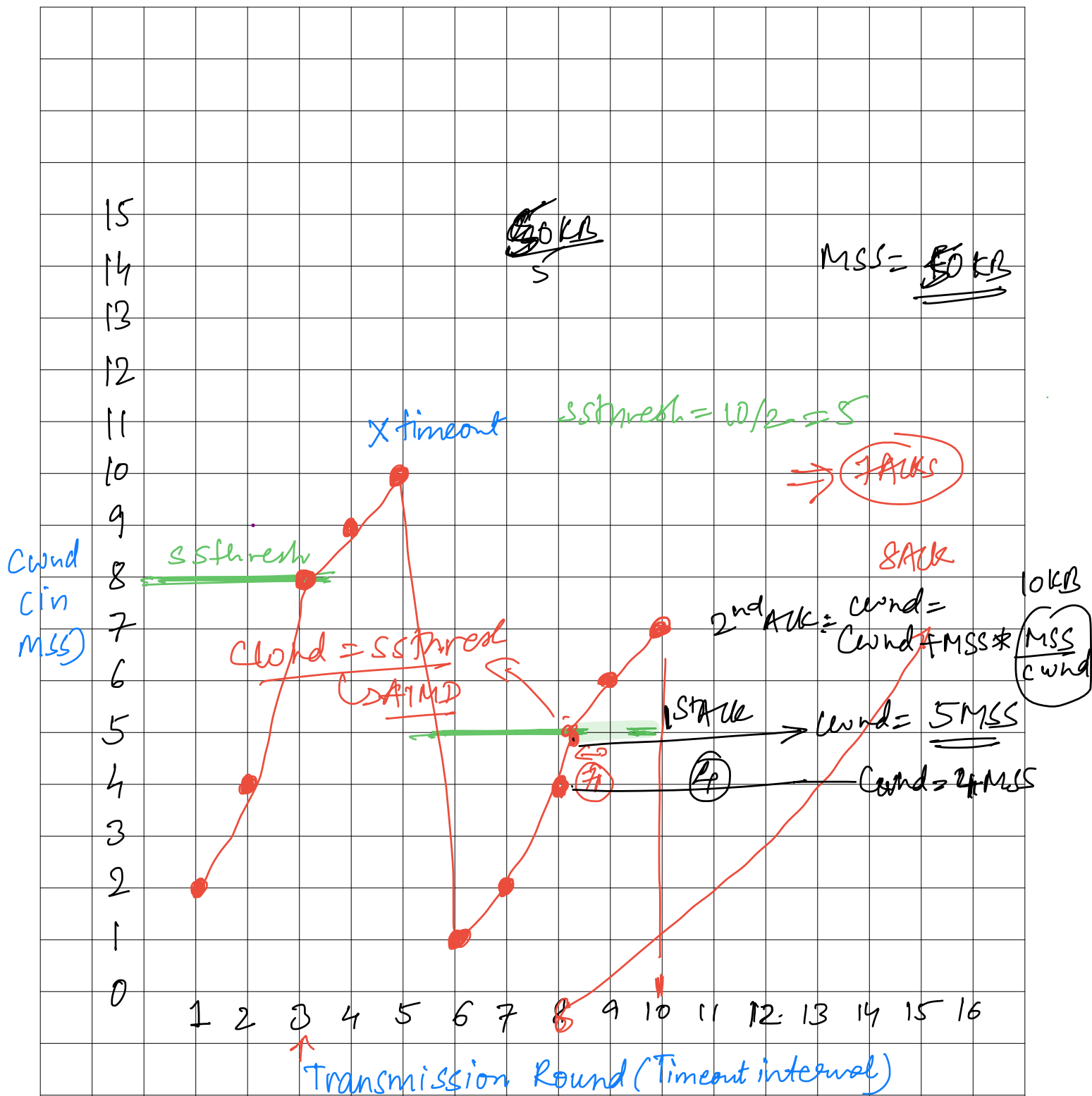


$Cwnd = 1MSS$

$S_1, S_2, \dots, S_8 \Rightarrow$ sent ; S_1 is lost
 RX receives $S_2 \rightarrow$ Ack 1. $\rightarrow$ dup Ack ①
 $S_3 \rightarrow$ Ack 1 $\rightarrow$ ②
 $S_4 \rightarrow$ Ack 1 $\rightarrow$ ③

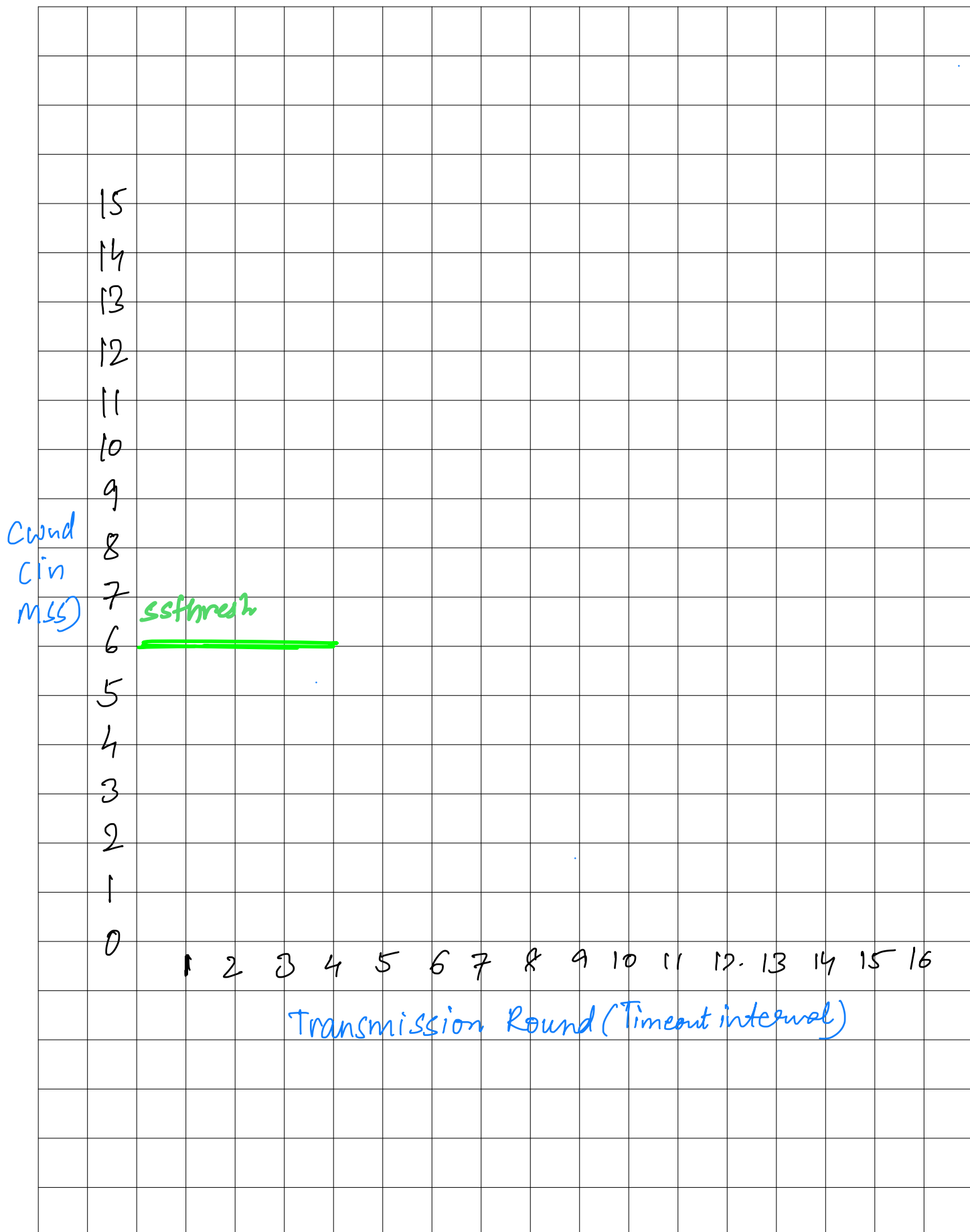


Q. 3 dup Acks during 6th transmission round
 Total 7 dup Acks received, after which a new Ack is recd



Q.1 Timeout occurs during 5th transmission. $cwnd_{\text{after trans}} = ?$

Q.2 Timeout occurs during the 8th & 14th transmission round / interval. Plot the cwnd value



If ACKs arrive on time $\Rightarrow$ double cwnd; if new cwnd $\leq$ ssthresh
 $cwnd = cwnd + 1MSS$; otherwise

If timer expires $\Rightarrow$ a) $ssthresh = \text{current cwnd} / 2$
 b) new cwnd = 1MSS

